Advanced Neural Engineering
脳工学特論

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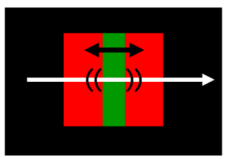

Ryohei Nakayama

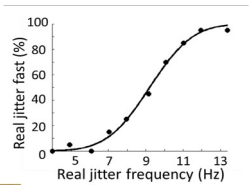

Kyuto Uno


Yu-Hsuan Tseng


Dongyang Yan

Perception, behavior, and cognition (psychophysical measurements)







Correlation

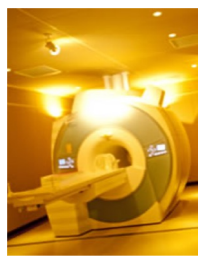
Causality

Brain activity / brain structure (MEG, EEG, MRI)

Magnetoencephalography (MEG)

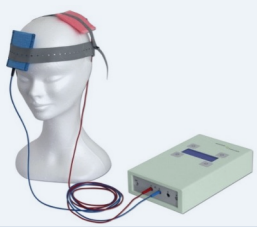


Magnetic resonance imaging (MRI)



Brain intervention: testing causal hypotheses

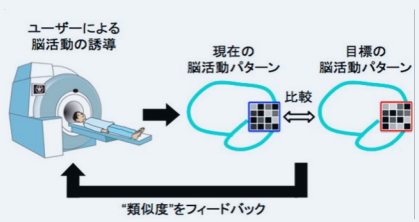
Transcranial electrical stimulation (tES)



Transcranial magnetic stimulation (TMS)



Neurofeedback

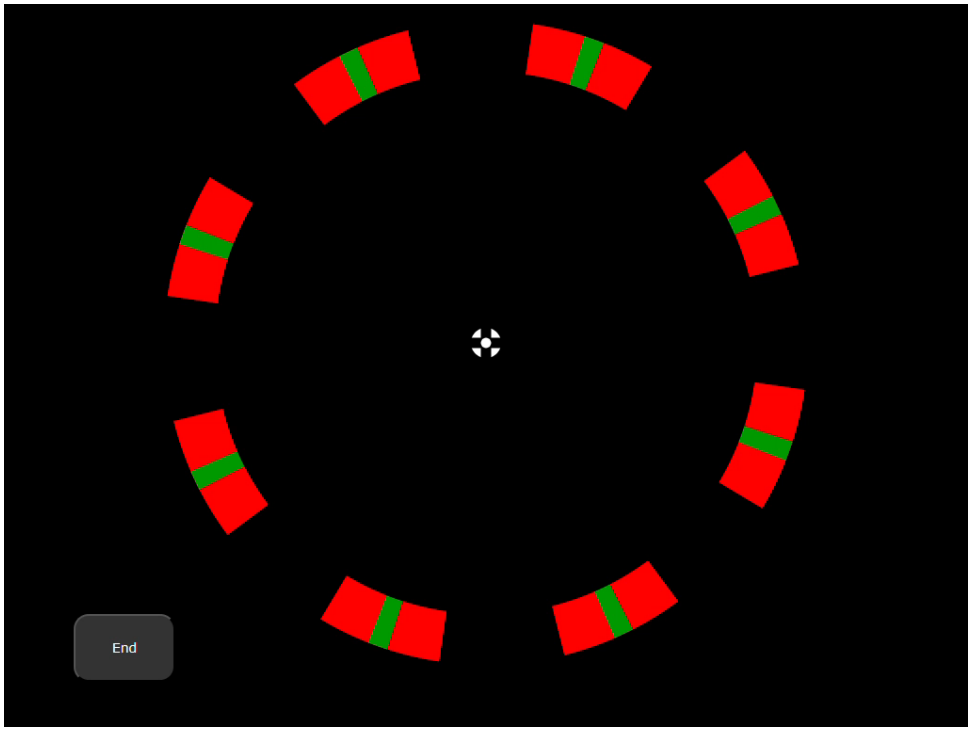


Research examples from our laboratory
（研究室の研究例）

- Neural oscillations and visual perception
- Perceptual tasks as indirect measures of brain state
- Understanding and inducing alcohol-like cognitive and brain states
- Encoding models for selecting images and sounds

Illusory jitter at around 10 Hz

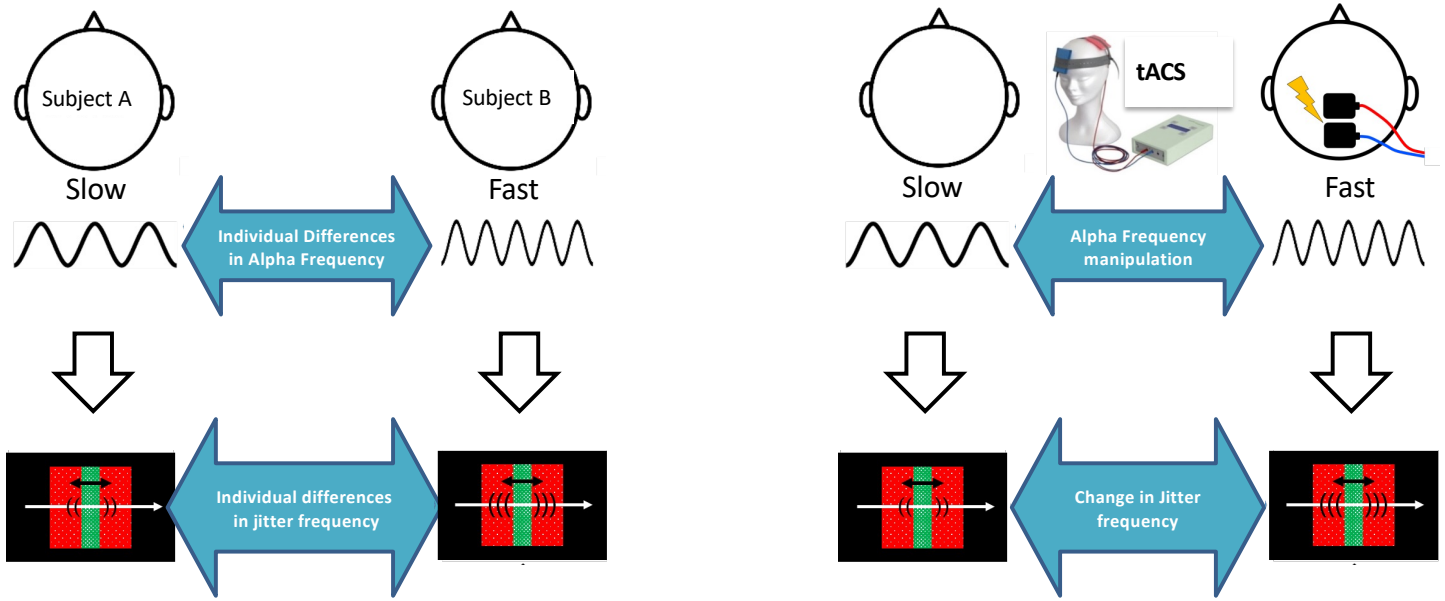
Keep looking at the center.



The green segments may appear to jitter despite smooth physical motion.

Illusory jitter perception and individual alpha frequency

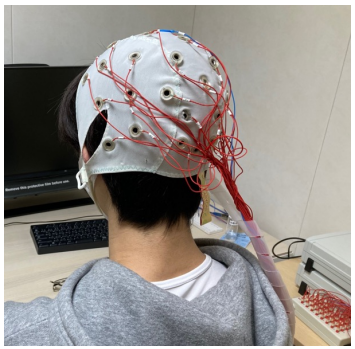
（ジター錯視の知覚と個人のアルファ周波数）



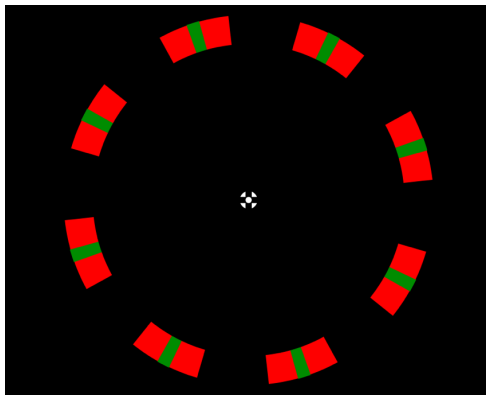
Minami & Amano, Current Biology (2017)

Estimating alpha frequency with a perceptual task

（ジター錯視測定によるアルファ波計測）



EEG/MEG require recording equipment and setup time.



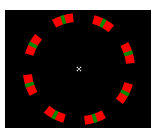
A perceptual task provides an indirect estimate of alpha frequency.

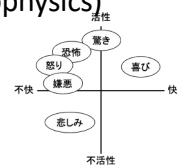
Research goal: relate everyday perceptual measures to cognition and changes in brain state.

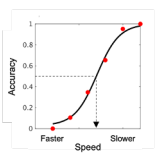
Understanding and inducing tipsy-like cognitive and brain states
（ほろ酔い時の認知・脳状態の理解と介入による類似状態の誘導）

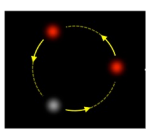
Research goal: characterize subjective, behavioral, and brain states during drinking

Perception, behavior, and cognition (psychophysics)







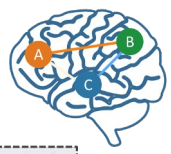


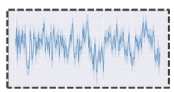
How does alcohol change perception, emotion, and behavior?

Optimal transport

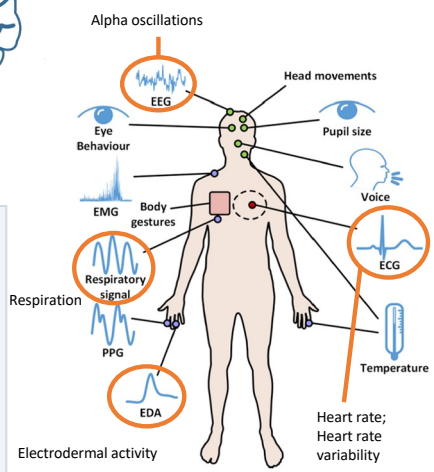


Resting-state / task fMRI Functional connectivity





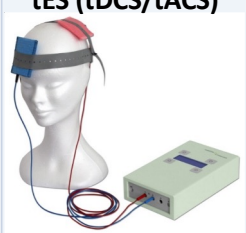
Physiological measures



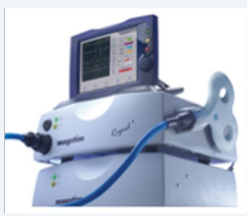
Giannakakis et al., 2019

Inducing a tipsy-like (ほろ酔い) state through brain intervention

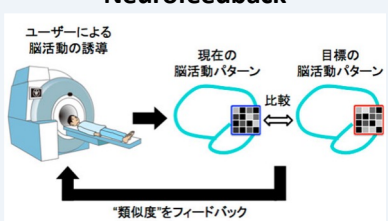
tES (tDCS/tACS)



TMS

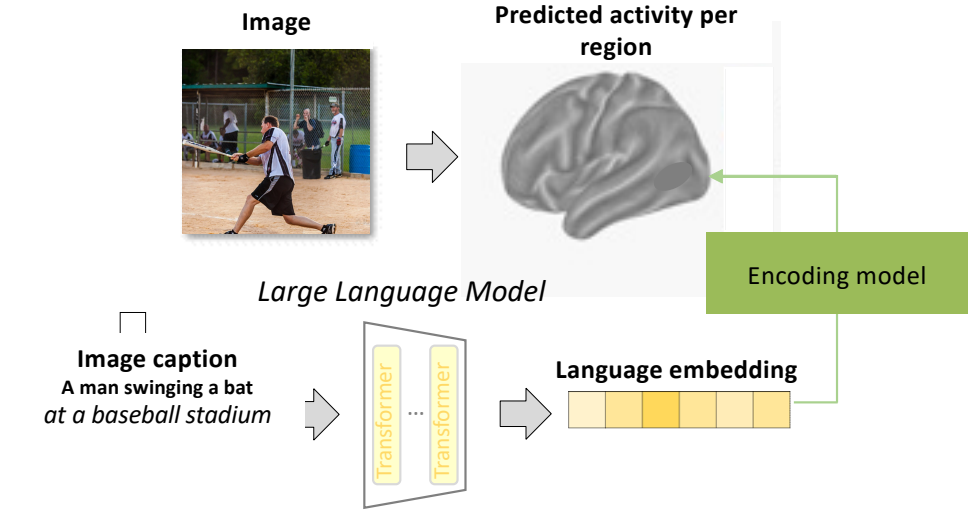


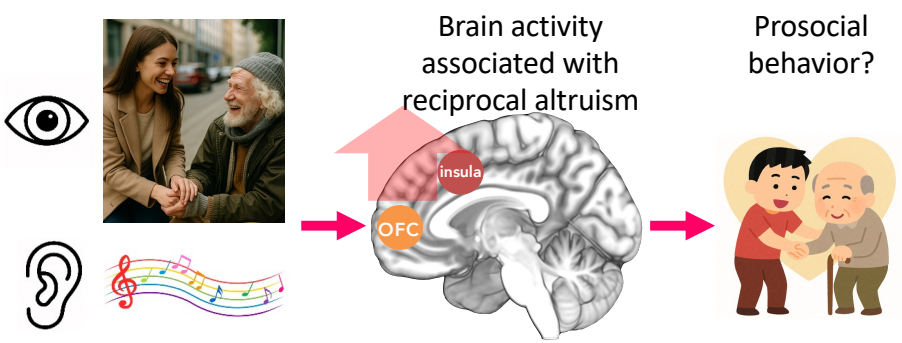
Neurofeedback



Optimizing Sensory Stimuli with Encoding Models to Modulate Mental State

（エンコーディングモデルを用いた感覚刺激の最適化による心的状態の変容）





- Predict brain responses to images and sounds with an encoding model
- Use the model to guide a generative model that synthesizes images and sounds predicted to enhance brain responses related to a target brain state
- Measure behavior to test whether the stimuli alter the target state

Course Goal and Use of generative AI

Goal

- Learn the fundamentals of neuroscience through lectures and hands-on exercises.

Use of generative AI

- We take advantage of rapid advances in AI to conduct experiments and analyses that were previously difficult to complete in a short time.
- We use vibe coding to build experiments and analyses pipelines for the exercises.
- You are encouraged to use generative AI to deepen your understanding of the lecture content.

Course outline | Part 1

Understanding perception: from behavior to brain

Lectures

Oct 6 Basics of psychophysics

Oct 13 Visual perception and functional brain mapping

Oct 20 Visual attention — Prof. Nakayama

Oct 27 Multisensory integration — Prof. Uno

Group exercise: quantify an illusion across conditions

Nov 10 Experimental design and short proposal presentations

Nov 17 Implement and pilot a psychophysical experiment

Nov 24 Collect data, analyze, and present results

Course outline | Part 2

Understanding neural representations with AI models

Lecture

Dec 8 Visual processing in brains and neural networks

Group exercise: texture bias of an AI model and its internal representations

Dec 15 Quantify AlexNet's texture bias with vibe coding

Then present a plan for a deeper analysis (1–2 slides per group).

Dec 22 Implement linear decoding or RSA; present results

Course outline | Part 3

Neural signals, decoding, and brain–machine interfaces

Lecture

Jan 5 Brain decoding, intervention, and BMI

Group exercise: design a neuroscience study or system

Jan 12 Develop the design, controls, and evaluation plan

Jan 19 Proposal presentations and discussion

Today's roadmap

- Thresholds and measurement procedures
  - Response bias: forced choice and signal detection theory
  - Comparing conditions in contrast sensitivity
- Comparative judgments: JND, PSE, and illusions
- Beyond thresholds: scaling and perceptual dynamics

Systems neuroscience needs quantitative behavior

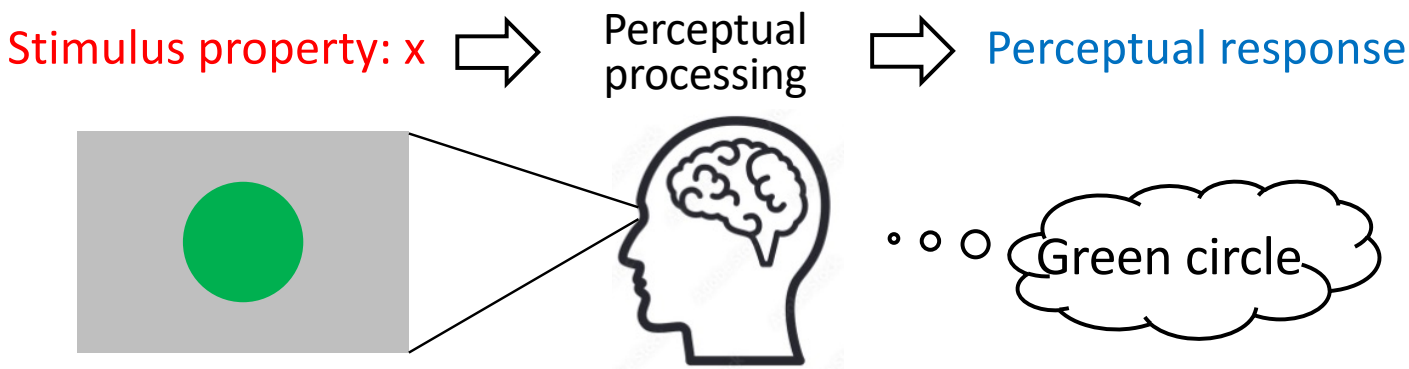
（システム神経科学には定量的な行動指標が必要）

Approach	Neural evidence	Behavioral evidence
Recording（記録）	Single-unit activity; EEG/MEG; fMRI	Relate activity to measured perception or performance
Lesions（損傷）	Location and extent of brain damage	Quantify impaired and preserved functions
Stimulation（刺激）	Controlled perturbation of neural activity	Test behavioral changes with appropriate controls

Psychophysics provides the behavioral measurements in all three approaches.

What is psychophysics?

（精神物理学/心理物理学とは）



- Relate controlled stimulus properties (x) to observable perceptual or behavioral responses (y).
- Quantify this relationship with a psychometric function, e.g., response probability as a function of x.
- Use the measured relationship to constrain hypotheses about perceptual processing.

Vision: from physical input to perception
（視覚：物理入力から知覚へ）



vs.

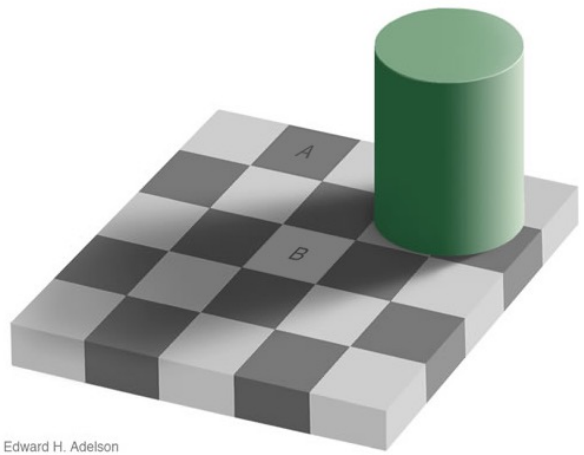


- A camera records light; perception depends on visual processing.
- Context can change appearance even when a local stimulus is unchanged.
- Visual illusions are ideal for revealing how human vision transforms physical input beyond what a camera records

Do A and B look equally light?

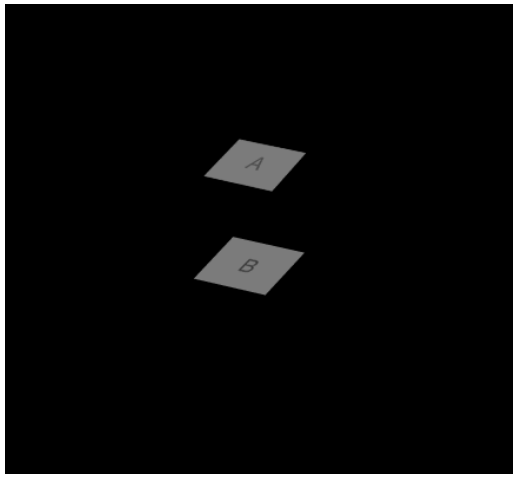
（AとBは同じ明るさに見えるか）

With the surrounding scene



Checker-shadow illusion: Edward H. Adelson.

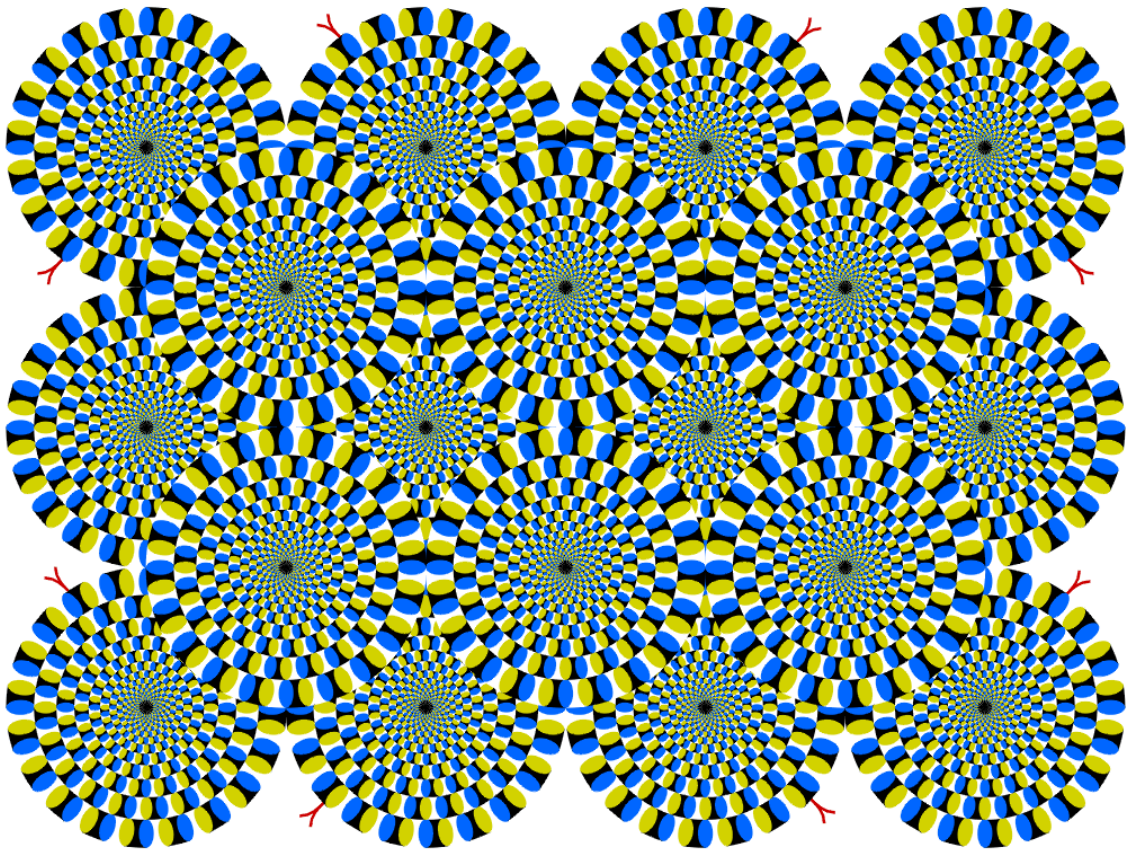
The same patches without the scene



The two patches have the same luminance, but their context changes their appearance.

Apparent motion in a static image

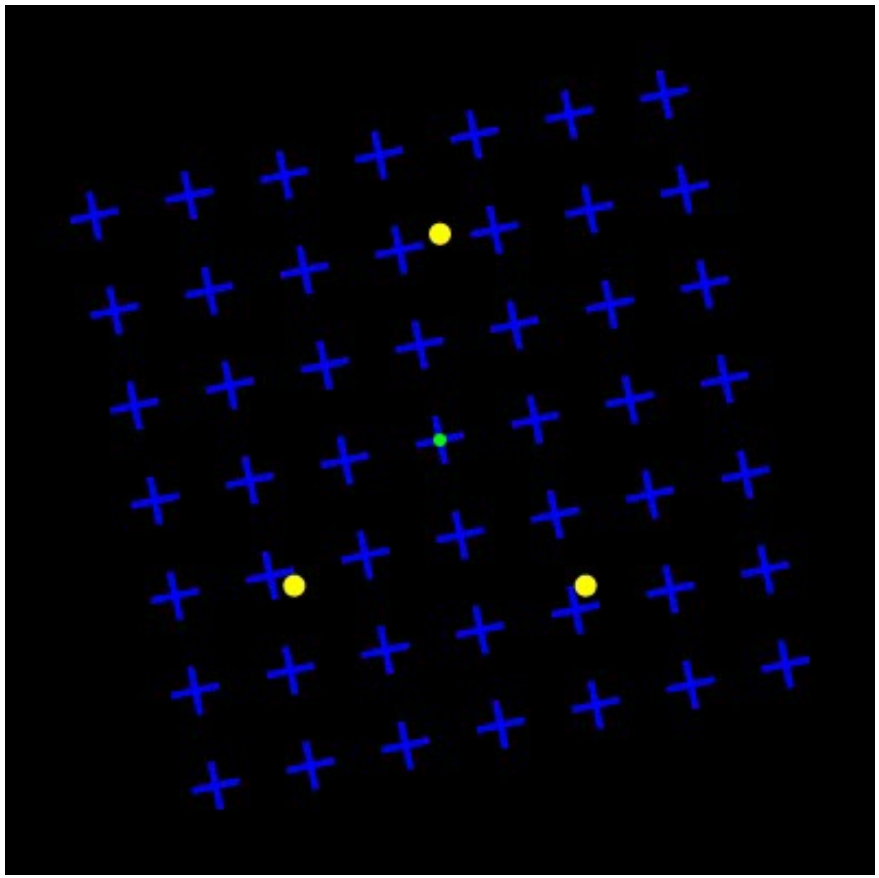
（運動しているように見える静止画）



Rotating Snakes illusion
(Kitaoka)

A visible target can disappear from awareness

（黄色の刺激が見えたり、見えなかったり）



Motion-induced
blindness
(Bonneh et al.)

Think first: how would you quantify an illusion?
（錯覚をどう定量化するか）

- Choose one: checker shadow, rotating snakes, or motion-induced blindness.
- What exactly is the illusion? Describe the effect in measurable terms.
- What would the participant report? (an observable response)
- What stimulus property would you manipulate?
- Which measure would you use to summarize the effect?

Three minutes individually, then compare ideas with a neighbor.

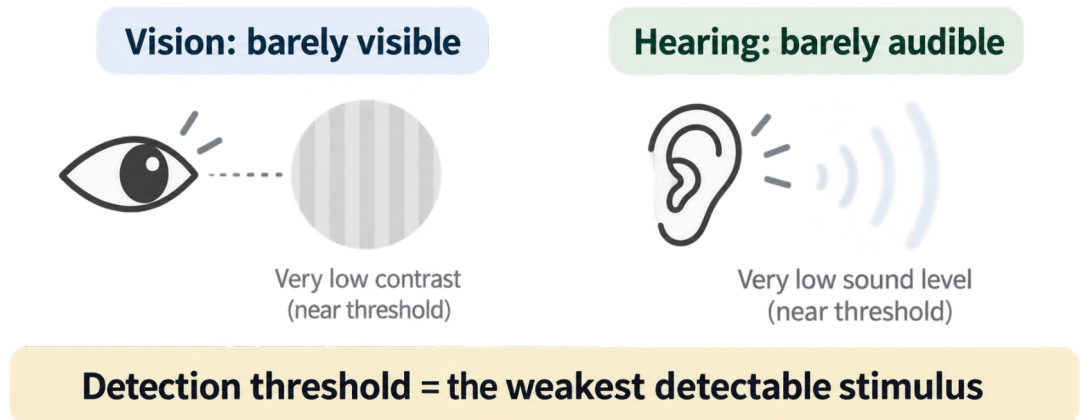
Choose the outcome that fits your question

（問いに応じて測定指標を選ぶ）

Question	Outcome
What is the weakest detectable stimulus or the smallest discriminable difference	Detection threshold / discrimination threshold (JND)
When are two stimuli perceived as the same?	Point of subjective equality (PSE)
How strong or what kind of percept is experienced?	Magnitude / impression / similarity
How does perception unfold in time (speed, rate, duration)?	Reaction time; event rate; percept duration

Detection and discrimination thresholds
（検出閾と弁別閾）

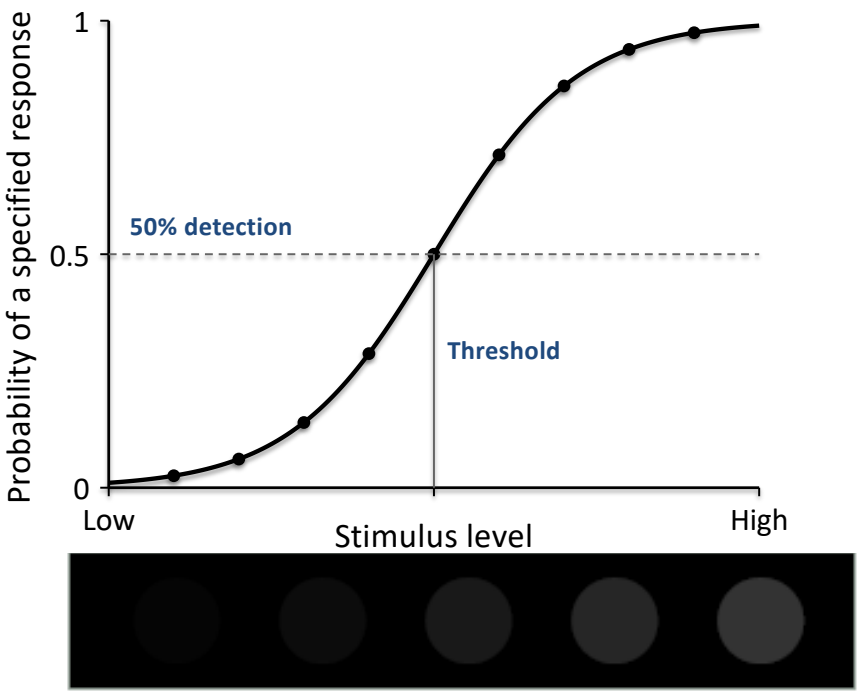
- A threshold is a stimulus level or difference at a specified response criterion.
  - Detection (absolute) threshold: how weak a stimulus can be detected.
  - Discrimination threshold / JND: how small a difference can be discriminated.
- A lower threshold indicates better sensitivity



Procedures for estimating thresholds
（閾値の測定法）

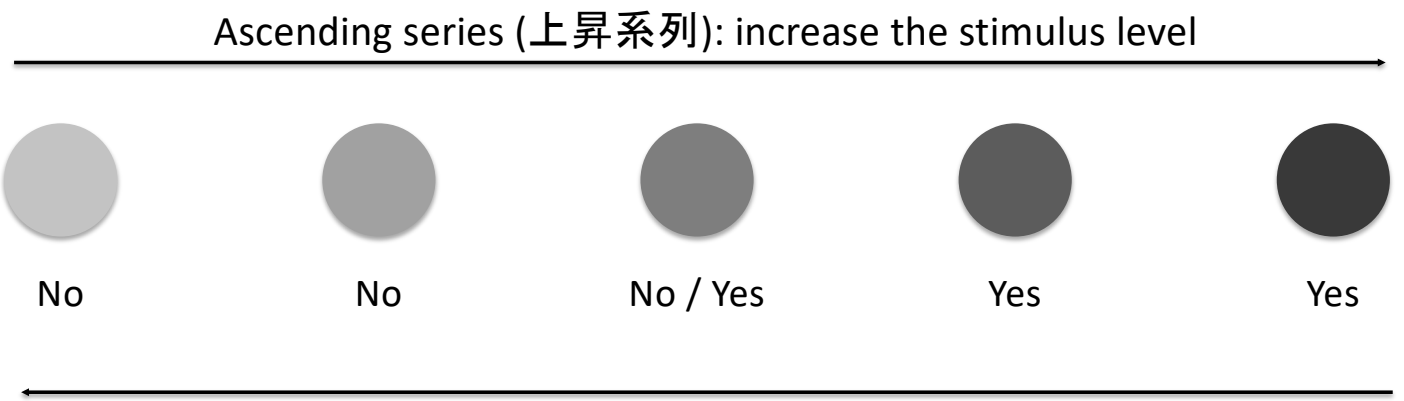
Procedure	How stimulus values are selected	Main consideration
Constant stimuli （恒常法）	Repeat preset levels in randomized order	Samples the function across a range
Limits （極限法）	Run ascending and descending series	Predictable order can influence responses
Staircase （階段法）	Adapt the next level to previous responses	Efficient near its target; the rule matters
Adjustment （調整法）	The observer directly adjusts a stimulus	Quick matching; depends on the setting strategy

Method of constant stimuli
（恒常法）



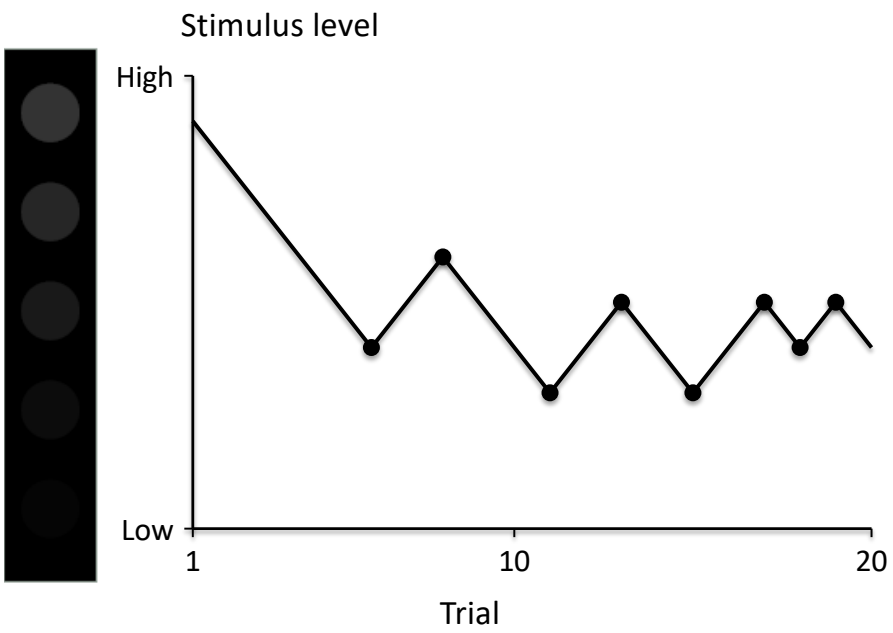
- **Participant task: report whether the stimulus was seen (Yes / No).**
- Select stimulus levels that span the response transition.
- Repeat each level multiple times; randomize trial order.
- Record each trial response, then fit a psychometric function (e.g. cumulative gaussian).
- Read the threshold at a predefined response criterion. The appropriate criterion depends on the task (explained later).

Method of limits
（極限法）



- **Participant task: report whether the stimulus was seen (Yes / No).**
- Run ascending and descending series; locate each response transition in each series.
- Average the transition estimates across repeated ascending and descending series to obtain the threshold.

Staircase method
（階段法）

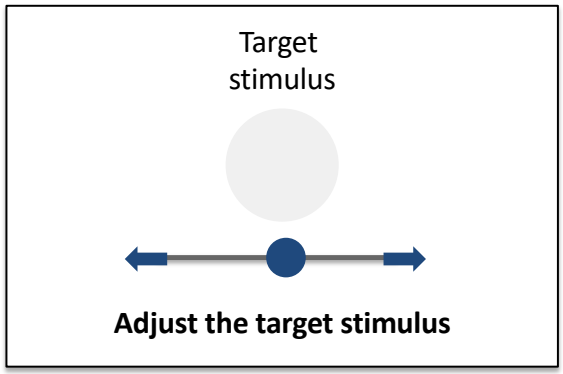


- **Participant task: report whether the stimulus was seen (Yes / No).**
- The next stimulus level depends on the previous response.
- No → increase stimulus strength;
  Yes → decrease stimulus strength.
- A 1-up / 1-down staircase converges to the level producing about 50% “Yes” responses.
- Near this level, the staircase repeatedly reverses direction.
- Estimate the threshold from the average of later reversals.

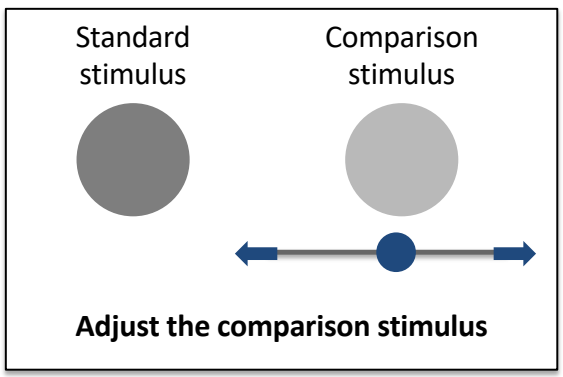
Method of adjustment

（調整法）

Detection threshold measurement



PSE measurement



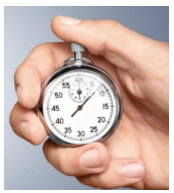
Participant task: directly adjust the stimulus until …

- Barely detectable → detection threshold
- Equal in appearance → point of subjective equality (PSE)

Repeat from different starting values; the mean setting estimates the threshold (or PSE).

Comparing the four procedures: measurement time and practical trade-offs

（4手法の比較：所要時間と実用性のトレードオフ）



Measurement time

Long



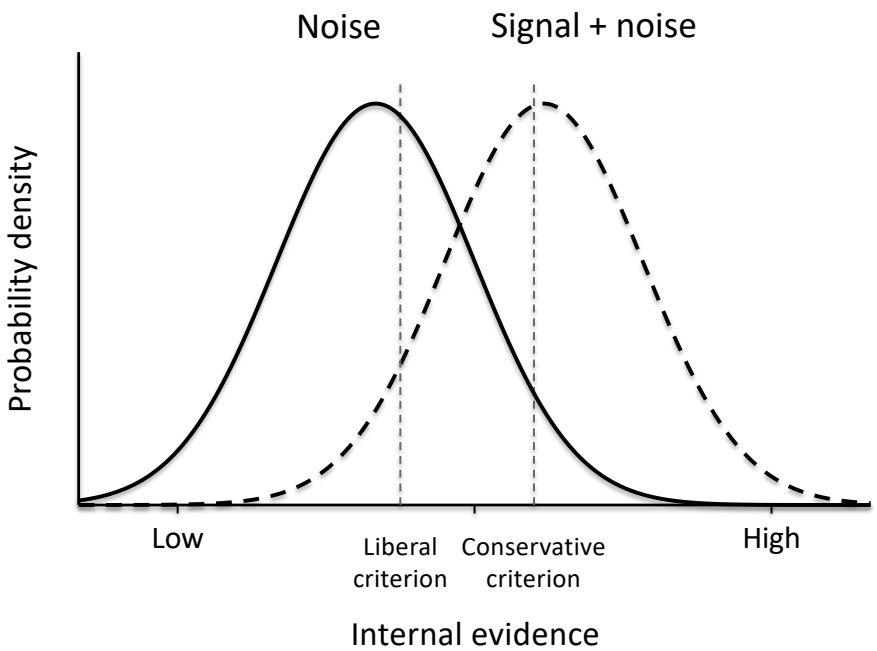
Short

- **Method of constant stimuli:**
  typically requires more trials because multiple stimulus levels are sampled repeatedly.
- **Method of limits, Staircase method:**
  can reduce measurement time; staircases concentrate trials near the target level.
- **Method of adjustment:**
  often the fastest, because the observer directly adjusts the stimulus.

Yes–no detection depends on response criterion

（イエス・ノー課題は反応基準に依存する）

Even at a fixed stimulus intensity, the internal sensory response (“internal evidence”) varies across trials. The dashed signal + noise curve shows this variability when a signal is present.

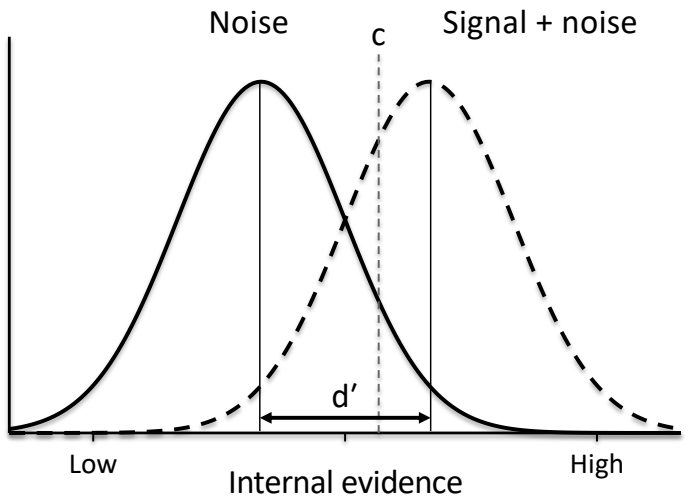


- The observer answers “yes” when this response exceeds a criterion (vertical dotted lines).
- For the same stimulus, a liberal (lower) criterion produces more “yes” responses than a conservative (higher) criterion.
- Now add the signal-absent (noise) distribution. “Yes” responses reveal criterion differences: more for liberal, fewer for conservative.

Signal detection theory: sensitivity and criterion

（信号検出理論：感度と基準）

	Yes response	**No response**
Signal present (signal + noise)	Hit	Miss
Signal absent (noise)	False alarm	Correct rejection



sensitivity d′ = z(H) – z(FA)

criterion c = −½ [z(H) + z(FA)]

- Signal detection theory separates sensitivity (d′) from criterion (c).
- With noise held constant, stronger signals generally produce larger d′ values.
- A detection threshold can be defined as the signal intensity needed to reach a chosen d′.

Same hit rate, different sensitivity

（ヒット率が同じでも感度は異なる）

	Participant A	Participant B
Hit rate	0.50	0.50
False-alarm rate	0.10	0.40
d’	1.28	0.25
Criterion c	+0.64 (conservative)	+0.13 (more liberal)

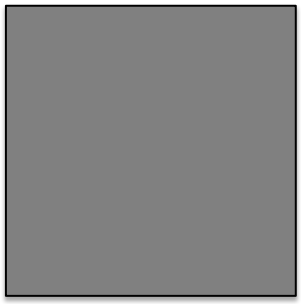
- **Under the 50%-hit-rate definition, both are at detection threshold.**
- **Yet B has lower sensitivity (d’) and a more liberal criterion (c) than A.**

Forced choice reduces yes–no criterion effects

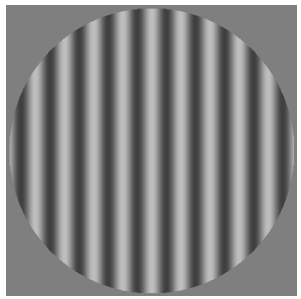
（強制選択法は反応基準の影響を減らす）

Two-alternative forced choice (2AFC): which interval/side contained the stimulus?

Interval 1



Interval 2



One interval/side contains the signal.

The other contains a blank.

Signal interval/side is randomized.

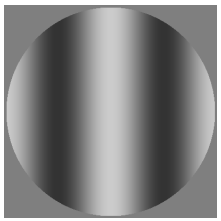
Chance accuracy = 50%; threshold is conventionally the level giving 75% correct (midway between chance and perfect).

What do contrast thresholds tell us about vision?

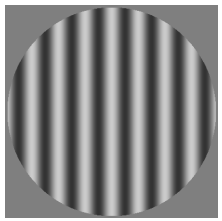
（コントラスト閾値から視覚の何がわかるか）

Question: does contrast sensitivity depend on spatial frequency?

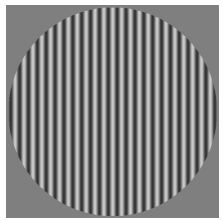
Across conditions: change spatial frequency f



1 cycle/degree



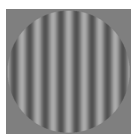
4 cycles/degree

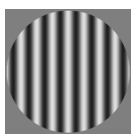


12 cycles/degree

Within each condition: vary contrast C to estimate a threshold







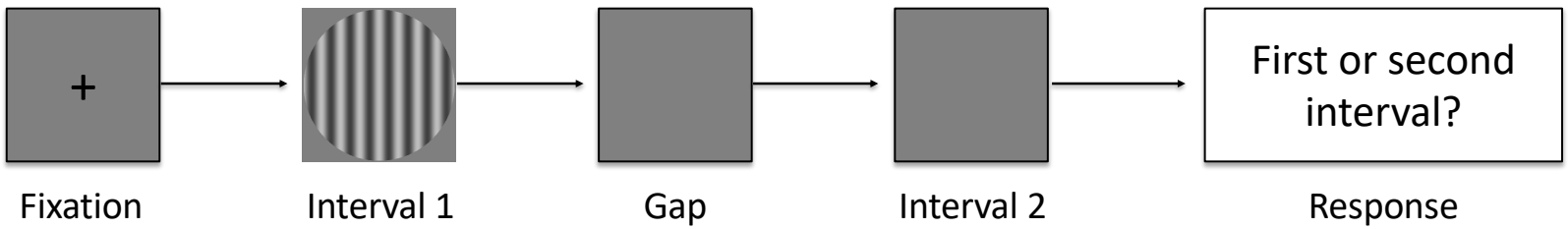
$$C = (L_{max} - L_{min}) / (L_{max} + L_{min})$$

Keep mean luminance, size, orientation, viewing distance, and duration constant.

Measuring contrast thresholds with 2AFC

（2AFCによるコントラスト閾値の測定）

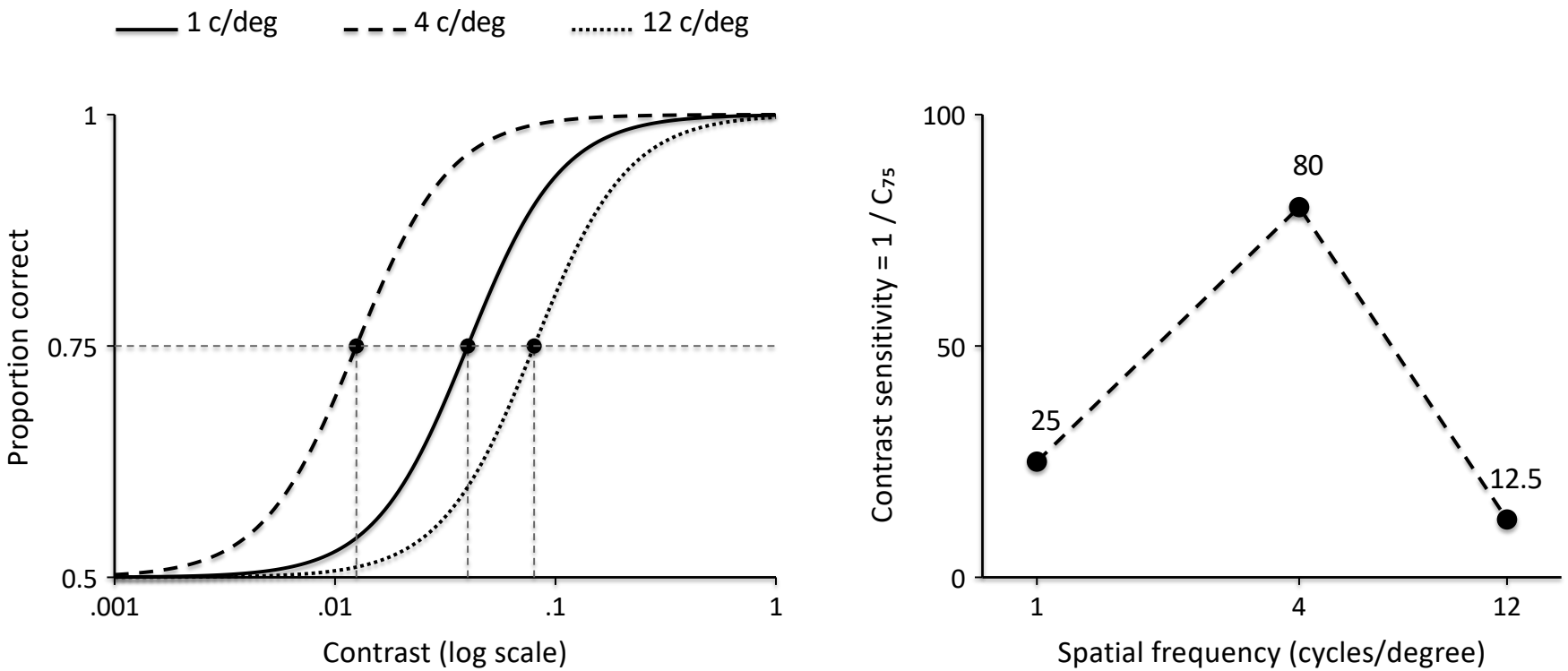
Two-alternative forced choice (2AFC): choose the interval containing the grating.



- At each spatial frequency, estimate the threshold using either procedure introduced earlier:
  - Constant stimuli: repeat preset contrast levels in random order.
  - Staircase: adapt contrast to correct/incorrect responses, using a rule that targets performance above chance.

From psychometric functions to contrast sensitivity

（心理測定関数からコントラスト感度へ）



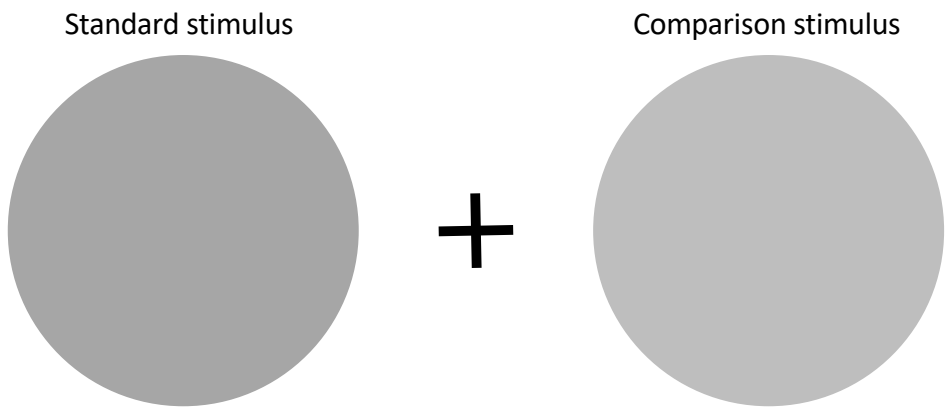
- **Contrast sensitivity is highest at intermediate spatial frequencies**
- **This tuning reveals a basic property of spatial processing in the visual system.**

Comparing two stimuli: the discrimination task

（二刺激の比較・弁別課題）

- Discrimination threshold (JND): the smallest difference between two stimuli that can be reliably told apart.
- Each trial shows a fixed standard and a variable comparison stimulus.
- The observer judges the comparison against the standard: e.g. “which circle is brighter?”
- Plot the proportion of “brighter” responses at each comparison value.
- The resulting curve is the psychometric function for comparison.

Which circle is brighter?



Discrimination threshold (just noticeable difference, JND) and PSE describe different aspects of a curve

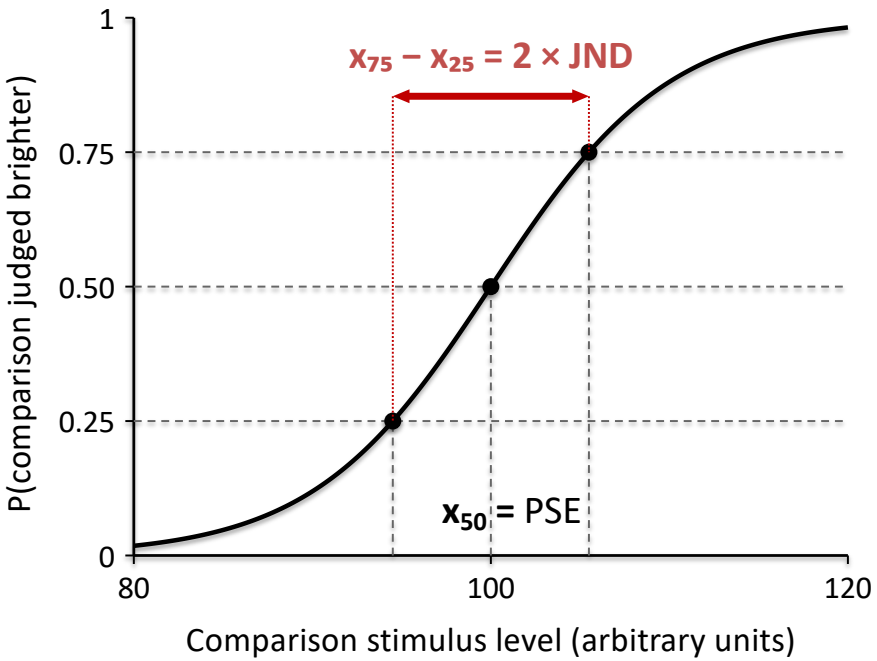
（弁別閾値（JND）と主観的等価点（PSE）は曲線の異なる側面を表す）

JND = (x_{75} − x_{25}) / 2

A conventional measure of the spread of comparative judgments.

Point of subjective equality (PSE) = x_{50}

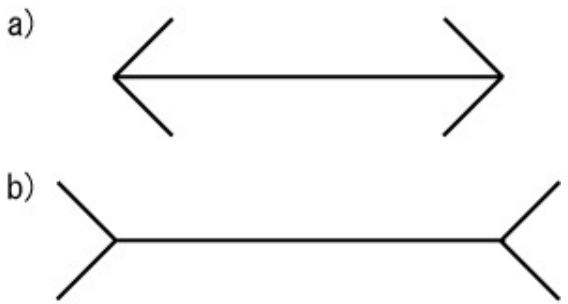
The comparison value judged brighter on half the trials.



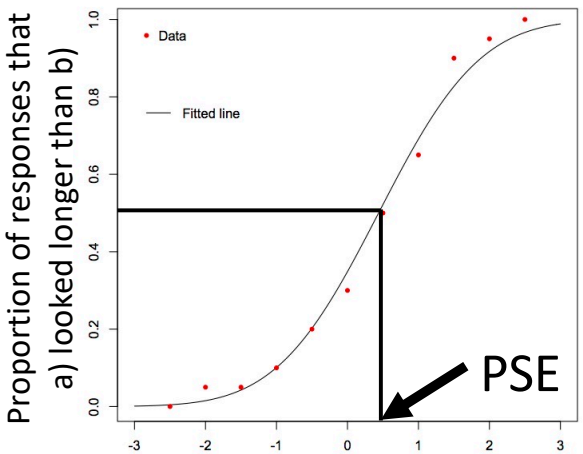
Quantifying an illusion with the PSE

（主観的等価点（PSE）による錯覚の定量化）

Müller-Lyer illusion

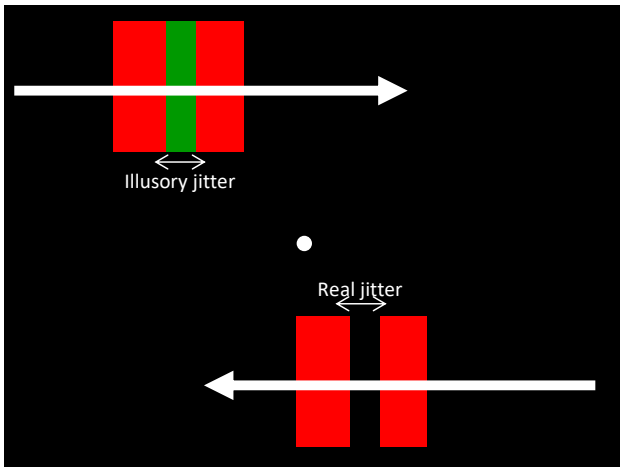


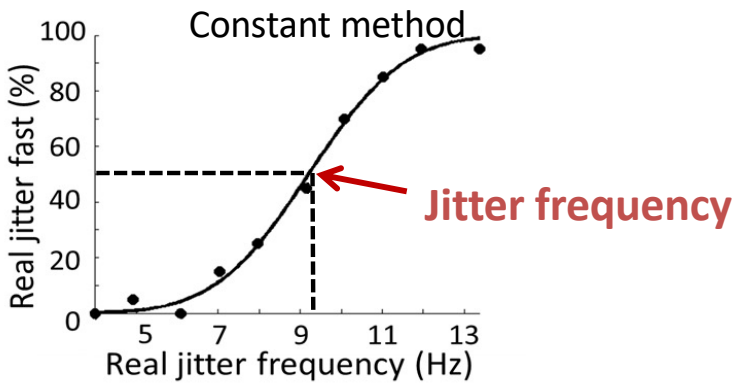
Fix the length of b) and vary the length of a)



- PSE is widely used to quantify illusions
- With the method of constant stimuli（恒常法）, the stimulus is varied over a fixed set of values and observers report which looks longer; the value yielding a 50% response is the PSE
- With the method of adjustment（調整法）, observers directly adjust the stimulus until it looks equal to the standard; the matched value is the PSE

Measuring the illusory jitter frequency（錯覚ジッター周波数の測定）

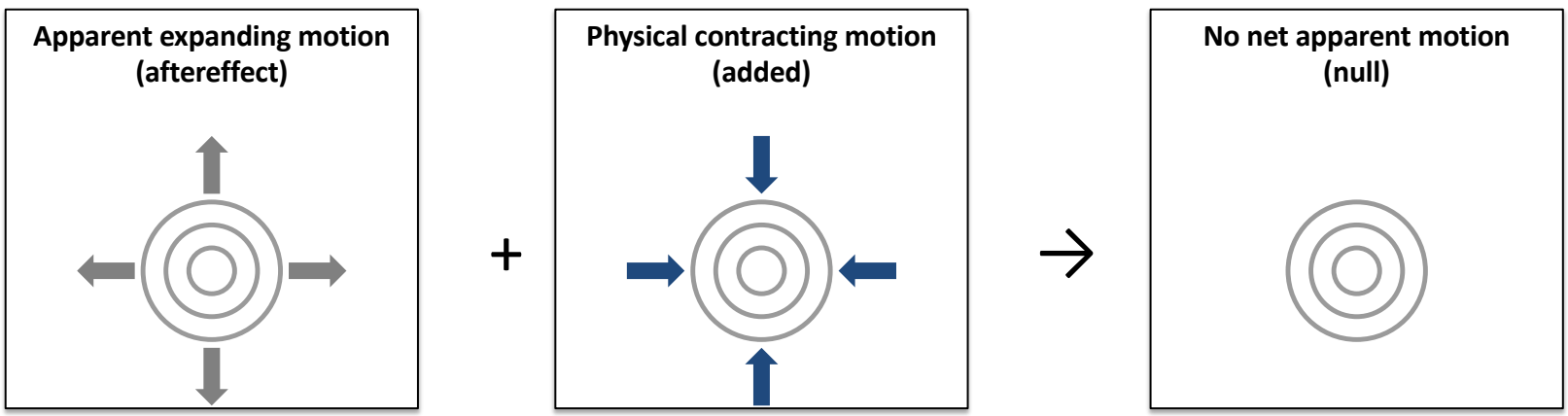




- Present the illusory jitter and a physical jitter stimulus
- With the method of constant stimuli（恒常法）, the physical jitter frequency is varied from 4 to 13 Hz and observers report which jitter looks faster; the frequency yielding a 50% response is the illusory jitter frequency
- With the method of adjustment（調整法）, observers directly adjust the physical jitter until it looks equal to the illusory jitter; the matched frequency is the illusory jitter frequency

Quantifying an illusion with nulling method
（ナリング法による錯覚の定量化）

https://illusion-forum.ilab.ntt.co.jp/motion-aftereffect/index.html



- Add an opposing physical change until the apparent effect is absent.
- The physical value at the null quantifies the perceptual bias.
- Nulling is a variant of the method of adjustment（調整法）: observers adjust the stimulus to a null rather than to a match.

Choose the outcome that fits your question

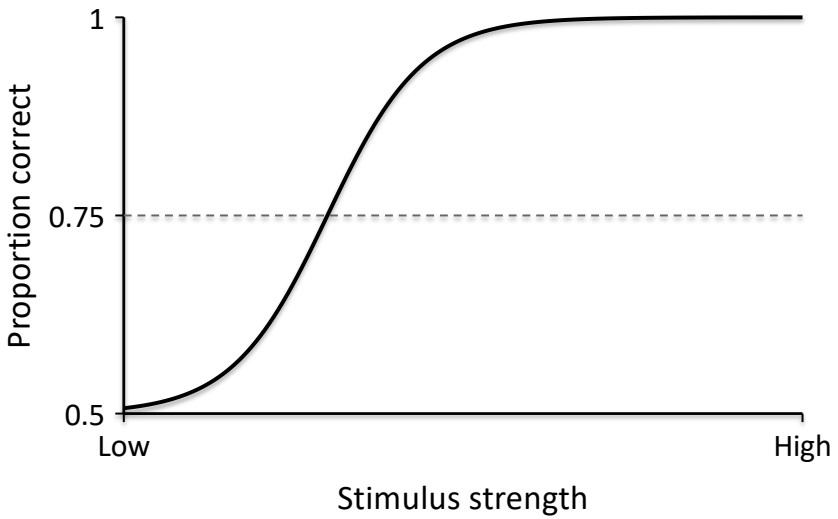
（問いに応じて測定指標を選ぶ）

Question	Outcome
What is the weakest detectable stimulus or the smallest discriminable difference	Detection threshold / discrimination threshold (JND)
When are two stimuli perceived as the same?	Point of subjective equality (PSE)
How strong or what kind of percept is experienced?	Magnitude / impression / similarity
How does perception unfold in time (speed, rate, duration)?	Reaction time; event rate; percept duration

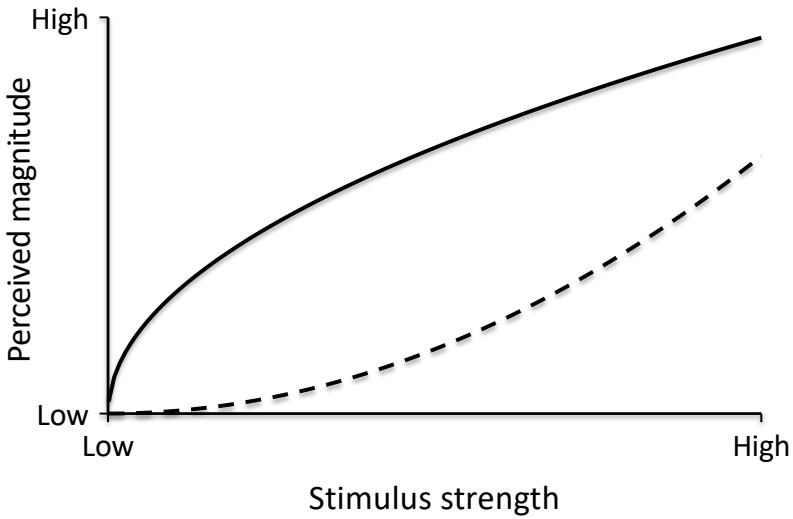
Detection threshold is not perceived magnitude

（検出閾と感覚の大きさは別物）

Detectability



Suprathreshold appearance



Equal detectability does not imply equal perceived intensity or quality, especially in the suprathreshold range.

Beyond threshold: magnitude, qualities, and similarity

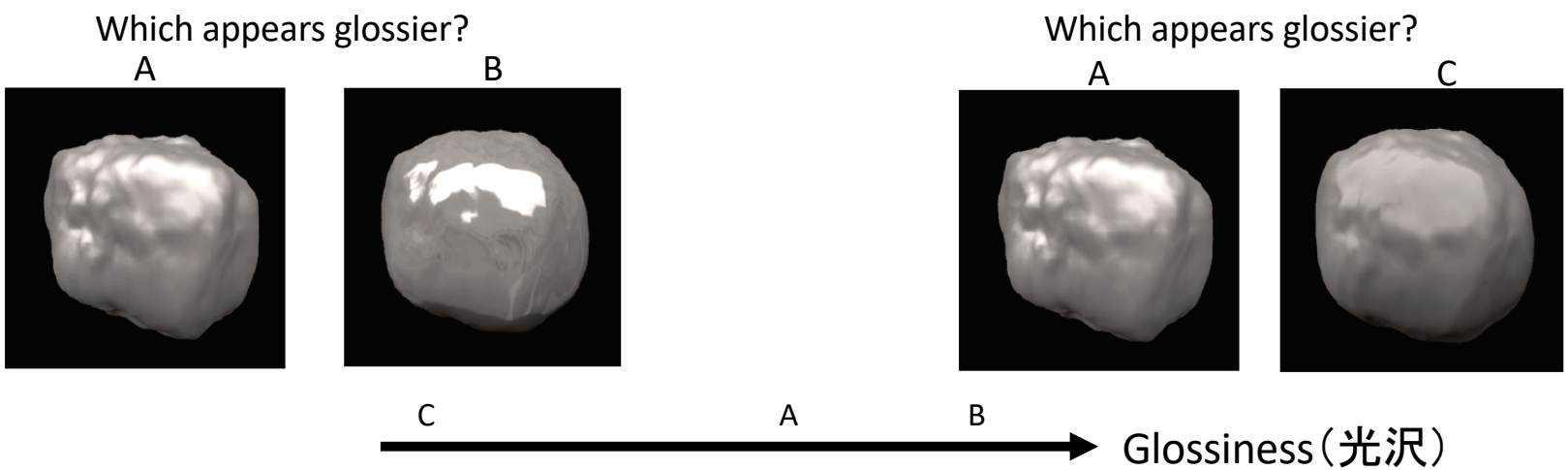
（閾値を超えて：感覚量・質・類似性）

Question	Outcome	Possible method
How much of a particular attribute is perceived? (e.g., brightness, glossiness, loudness)	Perceived magnitude	Paired comparison; Magnitude estimation
How is a stimulus perceived along different impression dimensions (e.g., natural–unnatural)?	Impression profile	Semantic differential (SD) method
How similar are two stimuli perceived to be?	Perceived similarity	Similarity judgement + Multidimensional scaling (MDS)

Perceived magnitude: indirect scaling by paired comparison

（知覚量：一対比較法による間接尺度構成）

- Compare two stimuli on a specified attribute (e.g., glossiness).
- Choice proportions constrain differences in perceived magnitude on a latent scale.
- Models such as Thurstone or Bradley–Terry specify the scaling assumptions.



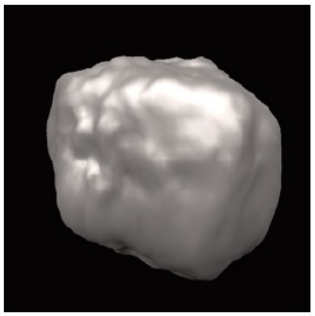
Illustrations: Takehiro Nagai
Basic Course on Vision and Imaging

Perceived magnitude: direct scaling by magnitude estimation

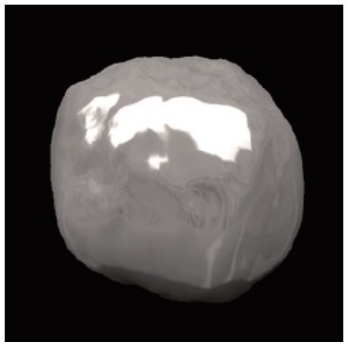
（知覚量：マグニチュード推定法による直接尺度構成）

- Assign numbers relative to a reference stimulus.
- If the reference is 10, a response of 20 indicates “twice as much.”
- Instructions, anchors, and response habits can affect the scale.

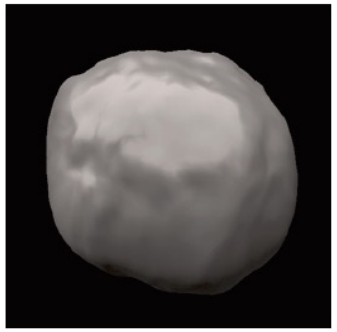
Reference glossiness: 10



Your rating of glossiness?



Your rating of glossiness?



Choosing a scaling method
（尺度構成法の選び方）

	Paired comparison	Magnitude estimation
Observer's task	Which stimulus has more of the target attribute?	Assign a number proportional to the perceived magnitude
Advantage	A simple comparative judgment	Directly yields a ratio-type scale with few trials
Cost / limitation	Trials grow with the number of pairs; a scale requires model assumptions	Sensitive to anchoring and to individual number-use habits
Interpretation	Relative positions on a fitted latent scale	Reported numbers are ratio-like, but comparisons across observers need normalization

Neither method is inherently better; the choice depends on the question.

Classical scaling laws: Fechner and Stevens
（古典的な尺度法則：フェヒナーとスティーヴンス）

Weber's law: ΔI / I = constant

- JND is roughly proportional to reference intensity (over a limited range).

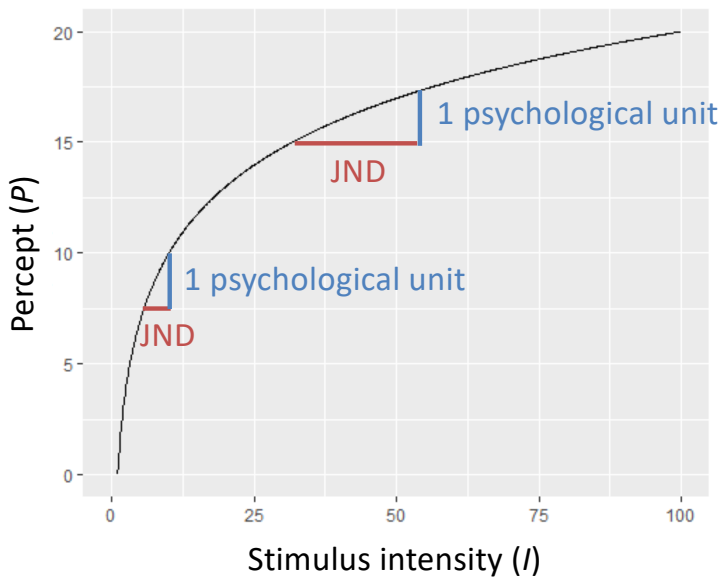
Fechner's law: $P = k \cdot log_{10} I$

- Equal JND steps = equal steps in psychological unit
- Relationship between stimulus intensity and perceived magnitude is logarithmic

Stevens: $P = k \cdot I^a$

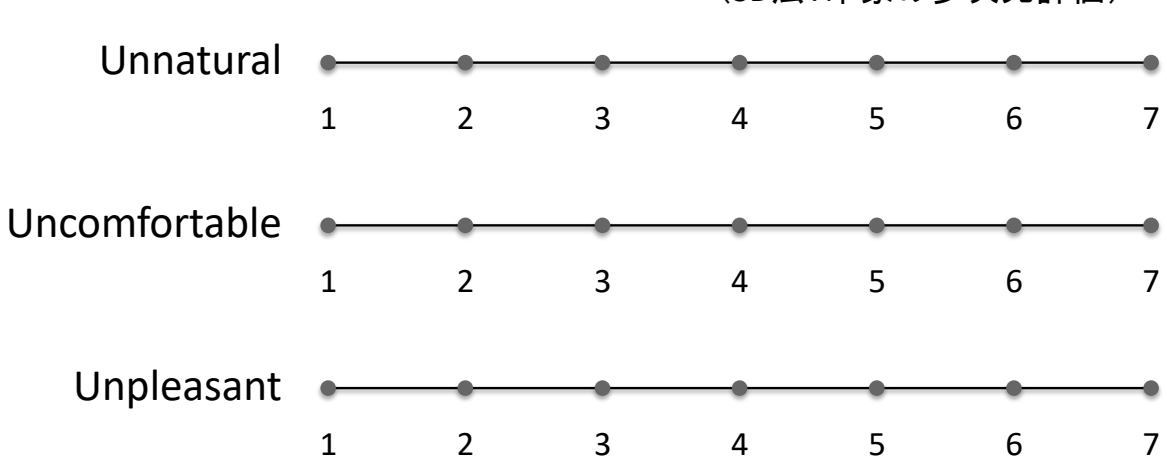
- Empirical power-law fit to magnitude-estimation data.

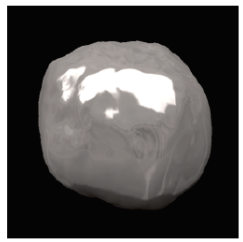
$$P = k \cdot \log_{10} I$$



Semantic differential (SD) method: profiling impressions

（SD法：印象の多次元評価）





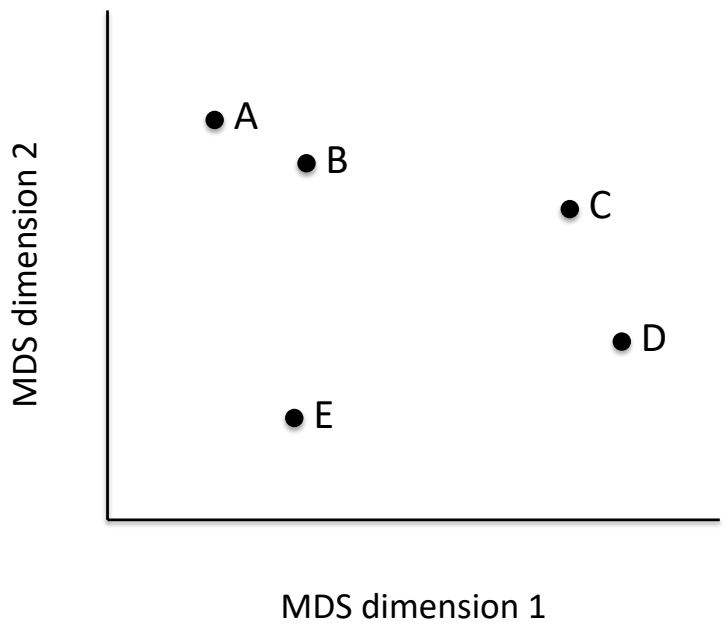


- Rate each stimulus on clearly defined bipolar adjective scales.
- Summarize ratings for each scale; factor analysis or PCA can reveal shared dimensions across scales.
- Widely used in applied evaluation, e.g., impressions of TV commercials or products.
- The same stimuli can also be assessed using magnitude estimation, but the two methods answer different questions.
- Magnitude estimation uses proportional numbers (e.g., 20 means twice the glossiness of a reference rated 10). The SD method uses fixed rating scales between opposite adjectives (e.g., unpleasant to pleasant).

Perceived similarity: multidimensional scaling (MDS)

（知覚の類似性：多次元尺度構成法（MDS））

- Represent pairwise relationships among stimuli as a (dis)similarity matrix.
- These relationships can be obtained
  - directly, by asking how similar A and B are
  - indirectly, by computing distances between SD rating profiles.
- Use multidimensional scaling to place stimuli so that the distances between them approximate these relationships.



Choose the outcome that fits your question

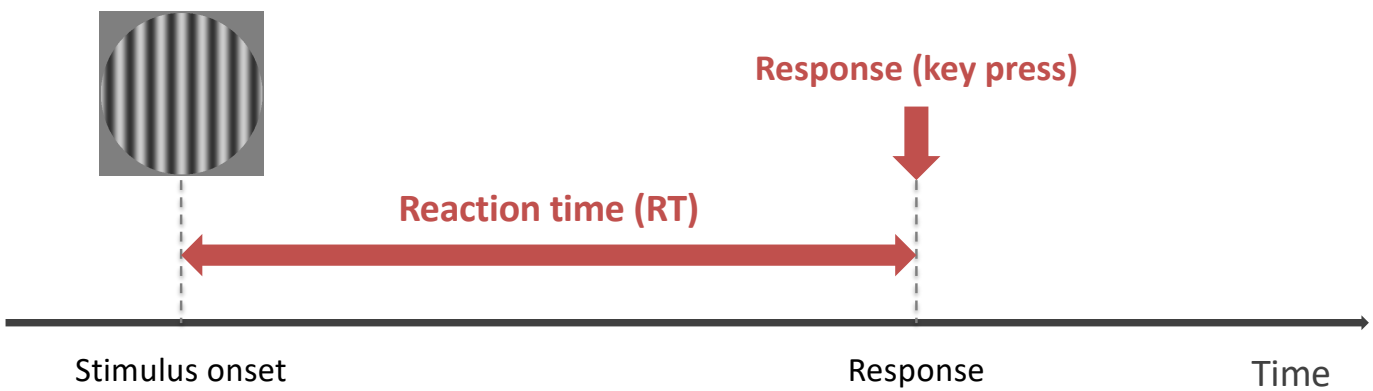
（問いに応じて測定指標を選ぶ）

Question	Outcome
What is the weakest detectable stimulus or the smallest discriminable difference	Detection threshold / discrimination threshold (JND)
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How strong or what kind of percept is experienced?	Magnitude / impression / similarity
How does perception unfold in time (speed, rate, duration)?	Reaction time; event rate; percept duration

Reaction time: stimulus-to-response latency

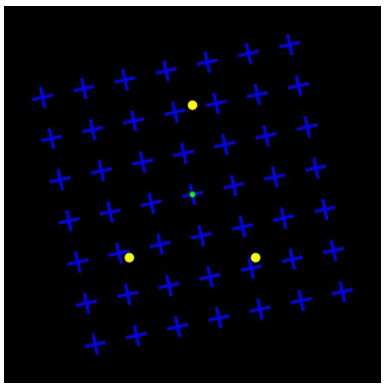
（反応時間：刺激から反応までの潜時）

- RT is the time from stimulus onset to the response.
- Simple RT uses one key for one stimulus (e.g., a grating). Choice RT uses different keys for different stimuli (e.g., vertical vs. horizontal gratings).
- RT reflects sensory, decision, and motor processes.
- Consider the speed–accuracy trade-off; report accuracy together with RT.
- Applicable from simple detection to more complex decisions, such as categorization, recognition or economic choice.



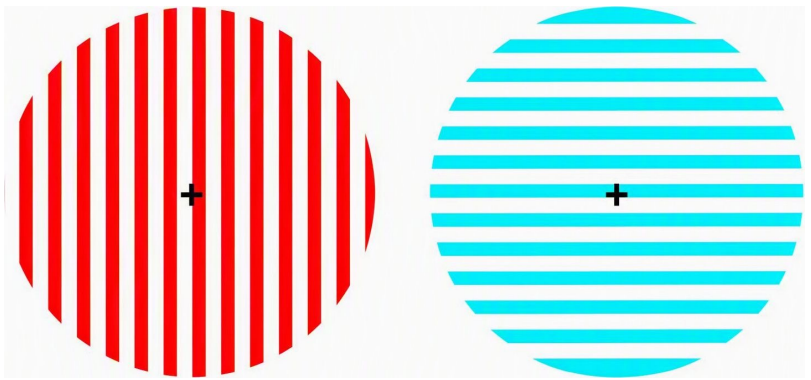
Perceptual dynamics: event rate and duration

Motion-induced blindness



- Report when the yellow target disappears and returns.
- Measure disappearance rate and duration.

Binocular rivalry



- Report the dominant or mixed percept.
- Measure switch rate and dominance duration.

Applications: adaptation and masking
（応用：順応とマスキング）

- Adaptation: measure how prior stimulation changes a threshold or PSE.
  - Vary adaptor–test similarity to test selectivity for attributes such as orientation or spatial frequency.
- Masking: measure how added noise or a mask changes performance.
  - Vary mask–test similarity to test selectivity for the same kinds of attributes.
- These effects constrain mechanisms; they do not uniquely identify a neural population.

Today's take-home messages

- Choose the outcome, response task, and stimulus-selection procedure separately.
- Detection threshold = the smallest detectable stimulus;
  Discrimination threshold (JND) = the smallest detectable difference.
- Estimate either one with constant stimuli, limits, staircase, or adjustment, reading the threshold at a stated criterion.
- A threshold gives the intensity or difference needed for a specified performance level. A PSE locates subjective equality.
- Forced-choice designs (e.g., 2IFC) reduce some biases; SDT separates sensitivity from criterion.
- How thresholds vary with stimulus parameters, such as spatial frequency, provides clues to the underlying perceptual mechanisms.
- Use magnitude, impression, similarity, or temporal measures when the goal is to describe the strength or quality of an experience rather than to locate a threshold.